

Part 4: SQL Validation – Explanation

Query 1: Return all employees and their departments

This query selects the name and department columns from the employees table. It provides a simple list of employees along with their department assignments.

Query 2: Return employees who have completed all assigned trainings

This query joins the employees table with employee_trainings and filters rows where the training status is 'completed'. It identifies employees who have successfully completed their assigned trainings.

Query 3: Count trainings per employee

This query counts the number of training records per employee by grouping on employee_id. It helps track training workload distribution across employees.

Query 4: Certifications expiring within 30 days

This query checks the expiry_date field in the certifications table and compares it with the current date plus 30 days. It flags certifications that are about to expire soon, ensuring compliance tracking.

Query 5: Employees with overdue trainings

This query identifies employees whose training due dates have passed (due_date < CURRENT_DATE) but whose status is not 'completed'. It highlights overdue training risks.

Query 6: Top 5 employees by completed trainings

This query counts completed trainings per employee, orders them in descending order, and limits the result to the top 5. It identifies the most compliant or high-performing employees in terms of training completion.